

Omid Erfanmanesh

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Summary

AI Engineer with 4+ years of experience building and deploying production AI systems across Generative AI, RAG, AI agents, computer vision and data engineering. Experienced in taking AI solutions from proof of concept to production on Azure, with applications spanning engineering knowledge systems, industrial computer vision and modern data platforms. Strong focus on reliable, maintainable AI systems that deliver measurable value in real-world environments.

Work experience

AI Engineer, Brembo S.p.A.

Jul 2026 – Present

- Building AI solutions for engineering and manufacturing teams, with a current focus on Generative AI and modern data platforms.
- Developing data pipelines and streaming applications on Azure Databricks using Apache Spark, Delta Lake and Event Hubs.
- Designing LLM applications and AI agents to automate internal workflows and improve access to engineering knowledge.
- Working with software, data and domain experts to turn business requirements into production-ready AI solutions.
- Contributing to the design of scalable data architectures using Unity Catalog and Azure services.

AI Engineer, Bosch VHIT S.p.A.

2022 – Jun 2026

- Shipped a RAG system on Azure used daily by 50 engineers and operators to search 1,000 pages of technical, code, and legal docs.
- Built an advanced retrieval pipeline combining hybrid BM25 + vector search, reranking, query rewriting and chunk augmentation, plus a multi-agent workflow for complex cross-document queries.
- Set up an evaluation loop with DeepEval, curated test sets, and user feedback, replacing ad-hoc releases with measurable iteration.
- Built an agentic workflow on Claude Sonnet that writes unit tests, opens PRs, and triggers CI, introducing automated testing in a codebase that had none.
- Built a document understanding pipeline (ingestion, OCR, LLM extraction, structured output) for invoice processing, removing manual data entry for the finance team.
- Pushed adoption of MLOps practices (Docker, CI/CD, model versioning, experiment tracking) across LLM and classical ML work.

- Deployed a CV model for scrap detection on the production line (~250–300 images/day), replacing a licensed solution and saving €12k.
- Deployed a CV model for crack detection on pump surfaces, replacing manual inspection on the line.

Data Scientist (Master's Thesis), Capgemini

2021 – 2022

- Trained a CNN for semantic segmentation to help UAVs identify safe emergency landing spots, then deployed it on a Jetson Nano, using pruning and quantization to fit the edge compute budget.
- Built training pipelines on AWS SageMaker and designed data cleaning and augmentation strategies to improve model robustness across varying lighting, terrain and weather conditions.

Projects

- Built a Telegram bot (live at [@ApplyGPTBot](#)) that answers student questions about Italian regional scholarships across all 20 regions, combining graph-based knowledge retrieval with a multi-agent setup and Tavily web search for things that change year to year.
- Split the work across specialized agents (one for scholarship lookup, one for regional eligibility, one for live web research), which cut down hallucinations and let answers draw on both the structured graph and current sources.
- Owned the full stack: data ingestion, graph construction, agent orchestration, and the Telegram interface.

Personal Project

Activities

DRAFT PoliTO: UAV Student Competition Team, Polytechnic University of Turin

2021 – 2022

- Developed AI components for UAV autonomy (trajectory planning, obstacle detection, and object recognition) using ROS, Gazebo, and ArduPilot.
- Worked alongside aerospace, software, and controls teammates to integrate perception and planning modules into simulated and real flight workflows.

Education

M.Sc. Data Science and Engineering, Polytechnic University of Turin

2019 – 2022

B.Sc. Information Technology Engineering, Jahrom University

2013 – 2018

Skills

GenAI & LLMs: RAG, Hybrid Retrieval, Reranking, Query Rewriting, Multi-Agent Systems, LangChain, Graph-based Retrieval, LLM Evaluation (DeepEval). Models: GPT-4 family (Azure OpenAI), Claude, Cohere embeddings.

ML & Data Science: PyTorch, TensorFlow, scikit-learn, Computer Vision, Anomaly Detection, Time-Series

Programming: Python (primary), C++, SQL, Bash, Structured Text (PLC)

Cloud & Data: Microsoft Azure (App Service, AI Search, OpenAI, Data Factory, Databricks, Synapse, ML), Spark, MySQL, MS SQL Server, MongoDB

MLOps & DevOps: Docker, Git, CI/CD, Model Versioning, Experiment Tracking, Model Monitoring

Certifications

Microsoft Certified: Azure Data Engineer Associate (DP-203)